

Business Sales Performance Analysis

1. Introduction

In today's data-driven business environment, organizations rely heavily on analytics to understand performance, identify growth opportunities, and make informed decisions. This project, Business Sales Performance Analysis, focuses on analyzing historical sales data to evaluate overall business health, profitability, product performance, and regional trends. The goal of the project is to deliver a client-ready dashboard and analysis report that can support strategic decision-making for business owners, startup founders, and analytics clients.

2. Objective of the Project

The primary objectives of this project are: - To analyze sales and profit performance over time - To identify top-performing and underperforming product categories and sub-categories - To evaluate regional and managerial performance - To assess the impact of returns on overall business performance - To generate clear business insights and actionable recommendations

3. Dataset Description

The analysis is based on a structured sales dataset consisting of three tables:

- Contains transactional sales data
 - Key fields include Order ID, Order Date, Ship Date, Customer details, Product Category, Sub-Category, Sales, Quantity, Discount, and Profit
 - Time period covered: 2021–2024
 - Regions to Regional Managers
 - Used to analyze manager-wise performance
 - Contains information on returned orders
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4. Data Cleaning and Preprocessing

Data cleaning and preprocessing were performed using Power BI Power Query to ensure accuracy, consistency, and analytical relevance.

Key Cleaning and preprocessing Steps:

- Removed non-business columns such as Row ID
 - Verified and corrected data types for dates and numeric fields
 - Removed duplicate records from the Returns table
 - Renamed columns for clarity and consistency
 - Created time-based features: Year, Month, Month Name, Quarter, and Year–Month
 - Derived business metrics such as Profit Margin
 - Created categorical fields like Profit Status (Loss, Medium Profit, High Profit)
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5. Key Performance Indicators (KPIs)

The following KPIs were defined to measure overall business performance: - Total Sales - Total Profit - Profit Margin (%) - Total Orders - Total Quantity Sold - Average Order Value (AOV) - Return Rate (%)

6. Analysis and Insights

6.1 Overall Business Performance

- **Total Sales:** 2.33M
- **Total Profit:** 292.30K
- **Profit Margin:** 12.56%
- **Average Order Value:** 455.20
- **Return Rate:** 5.79%

The business demonstrates strong revenue generation with healthy profitability and a controlled return rate.

6.2 Analysis

- Sales and profit show a consistent upward trend over the years
- Performance in 2023 and 2024 is significantly stronger compared to earlier years
- The trend indicates steady growth with no major volatility
- Technology is the top-performing category in terms of both sales and profit
- Office Supplies shows stable sales with good profitability
- Furniture has relatively high sales but a very low profit margin
- Top-selling products are primarily from the Technology category
- Top profit-generating products differ from top-selling products

- West and East regions contribute the highest sales and profit
 - Central and South regions show lower performance
 - Manager-wise analysis reveals significant performance differences across regions
 - Total Returned Orders: 296
 - Return Rate: 5.79%
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7. Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. **Focus on High-Margin Categories**
Increase investment and promotion in the Technology category to maximize profitability.
 2. **Improve Furniture Category Profitability**
Review pricing strategies, reduce excessive discounts, and optimize supply chain costs.
 3. **Regional Strategy Optimization**
Replicate successful strategies from West and East regions in Central and South regions.
 4. **Product-Level Optimization**
Promote high-profit products and bundle low-margin products with high-margin ones.
 5. **Reduce Returns Further**
Identify high-return products and improve quality checks, product descriptions, and customer expectations.
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8. Tools and Technologies Used

- **Power BI** – Data cleaning, modeling, visualization, and dashboard creation
 - **Power Query** – Data preprocessing and feature engineering
 - **Microsoft Excel** – Source data format
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9. Conclusion

This project successfully demonstrates the end-to-end analytics workflow, from data cleaning and preprocessing to dashboard development and business insight generation. The analysis highlights strong overall performance, clear revenue drivers, and specific

areas for improvement. The final dashboard and report are **client-ready** and showcase practical skills required for real-world data analytics roles.